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Histology: Histology of Connective Tissue, Muscle, Nerve, and Applications

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session objectives

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1. Describe and compare the general morphology, functions, and locations of cartilage, bone, and blood.
2. Compare and contrast the characteristics, function, and location of various types of muscle tissue.
3. Compare and contrast the function and morphology of various types of nervous tissue.
4. Explain histogenesis and development of different types of tissue.
5. Apply principle by identifying various tissue types.

Specialized Connective Tissue

- Specialized C.T.:
  1. Cartilage
     1. Hyaline
     2. Elastic
     3. Fibro
  2. Bone
     1. Spongy
     2. Compact
  3. Blood

Cartilage

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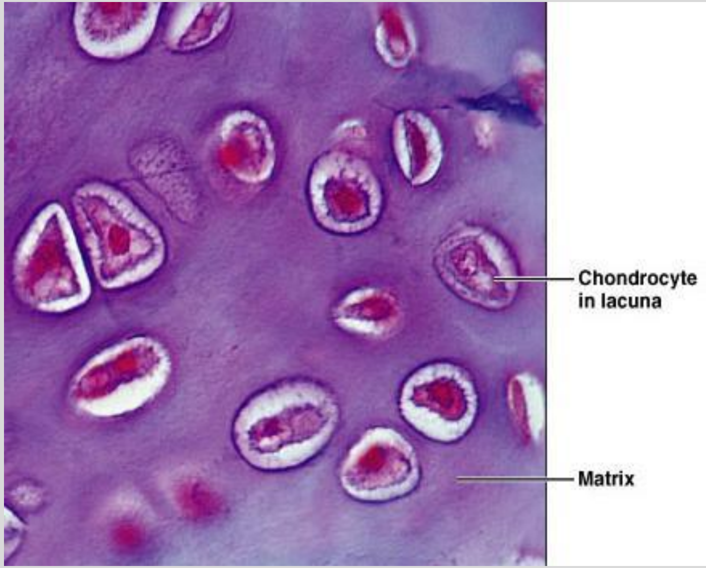
- Characteristics:
  a) Firm & durable (mechanical stress)
  b) Matrix 60-80% water
  c) Type II Collagen & elastic fibers
  d) High conc. of GAGs & proteoglycans
  e) Avascular (not innervated)
  f) Development/growth of long bones
  g) Chondrocytes in lacunae
  h) Covered perichondrium
- Types:
  1. Hyaline – most common
  2. Elastic – ears, larynx
  3. Fibrocartilage – vertebral discs, symphysis pubis, cranial sutures



Hyaline Cartilage

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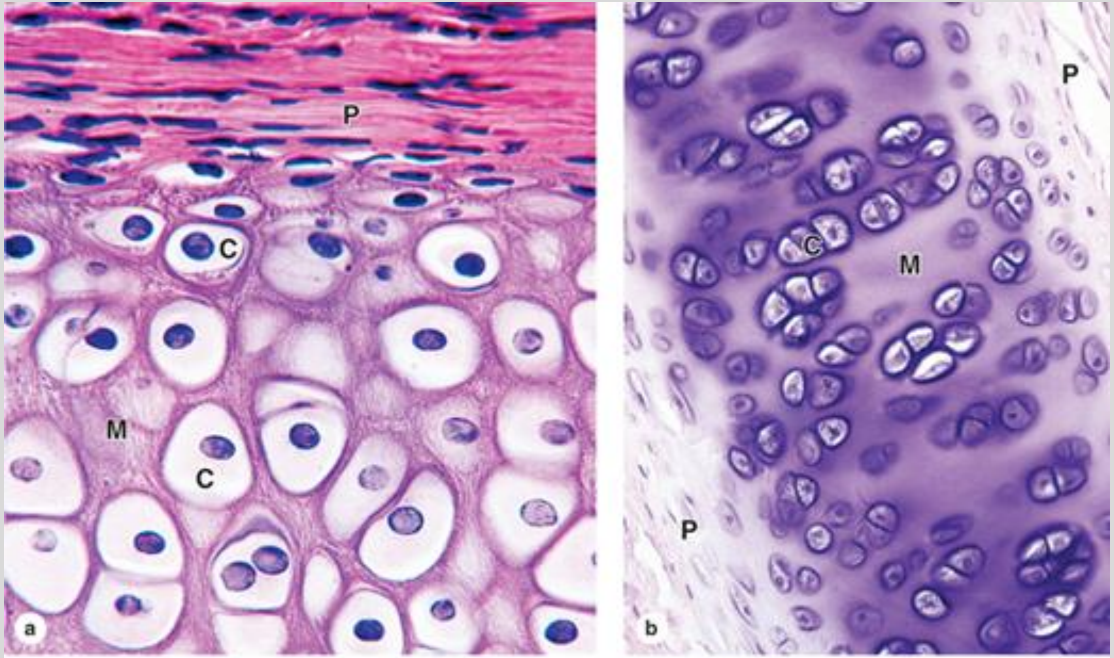
- Description
  a) Hyaline = glassy
  b) homogeneous & semitransparent
- Function
  a) Supports & reinforces
  b) Resilient cushion
  c) Resists repetitive stress
- Location
  a) Articular surfaces of movable joints
  b) Walls of larger respiratory passages
  c) Epiphyseal plates
- Loss → osteoarthritis
- Costal cartilage calcify with age
- Somatotropin (anterior pituitary)



Source: Anthony L. Mescher: Junqueira's Basic Histology: Text and Atlas, 15th Edition.

Elastic cartilage

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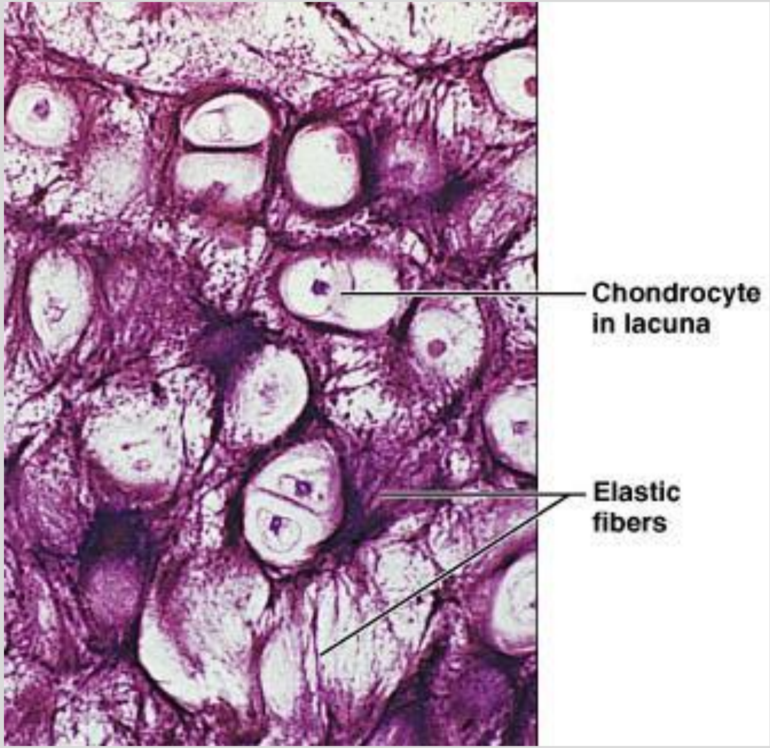
- Description
  a) Similar to hyaline cartilage + more elastic fibers
- Function
  a) Maintains shape of structure
  b) Allows great flexibility
- Location
  a) Supports external ear
  b) Epiglottis
  c) Upper respiratory tract



Fibrocartilage

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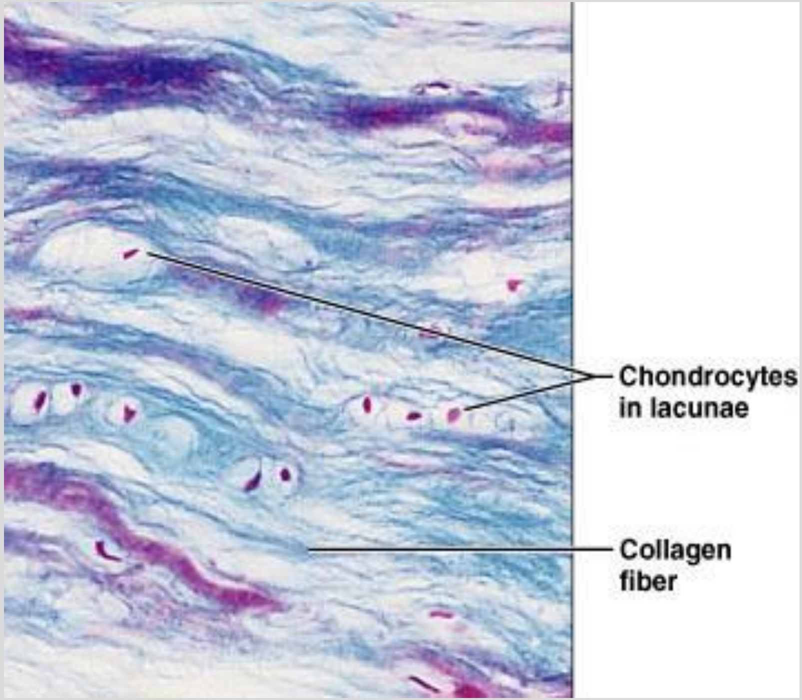
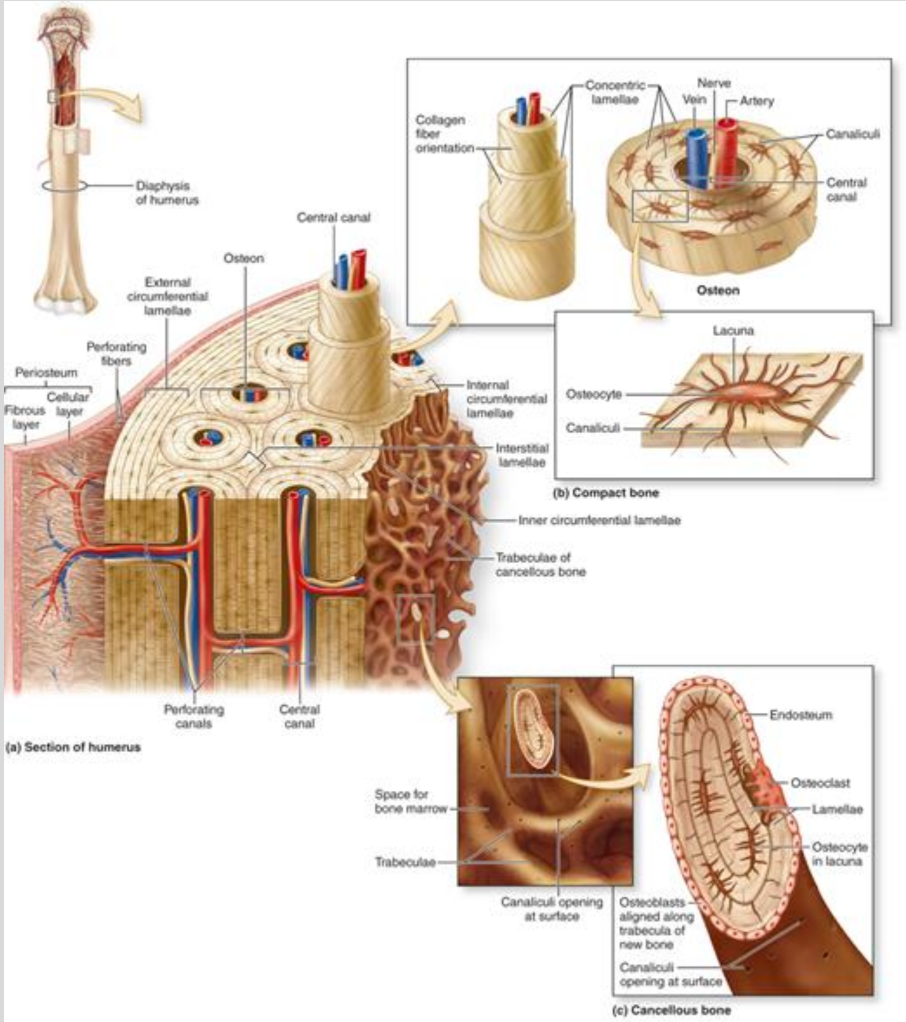
- Description
  1. Hyaline cartilage & dense connective tissue
- Function
  1. Tensile strength &ability to absorb compressive shock
- Location
  1. Intervertebral discs
  2. Pubic symphysis





Bone

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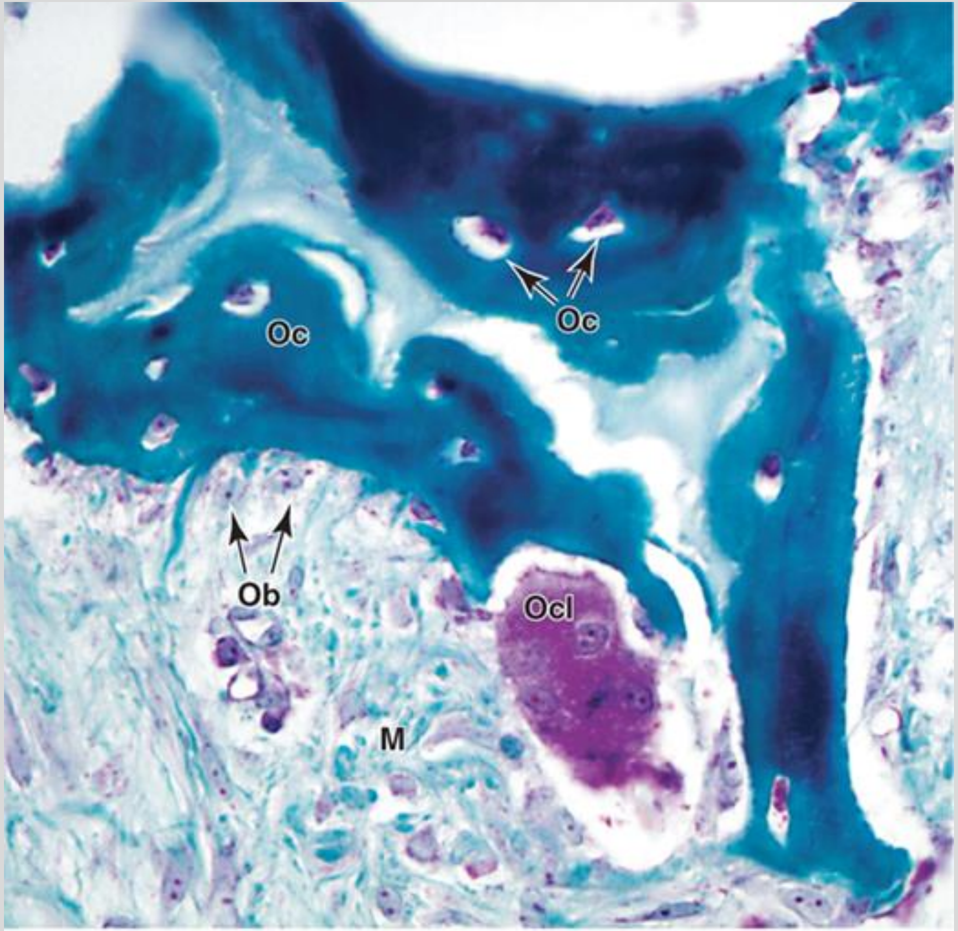
- Calcified matrix
- Function
  a. Supports & protects organs
  b. Provides levers & attachment site for muscles
  c. Stores Ca++ & other minerals
  d. Stores fat
  e. Red marrow (site for blood cell formation



Source: Anthony L. Mescher: Junqueira's Basic Histology: Text and Atlas, 15th Edition.

Bone cells

- 3 types of cells:
  1. Osteocytes
  2. Osteoblasts
  3. Osteoclasts

Bone types

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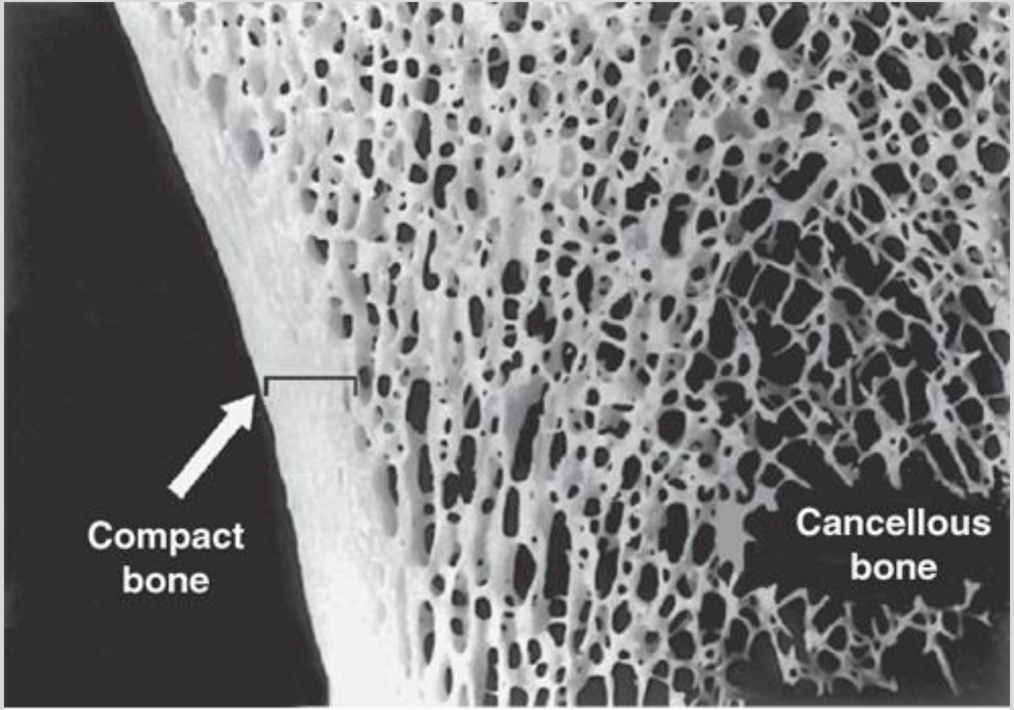

Source: Anthony L. Mescher: Junqueira's Basic Histology: Text and Atlas, 15th Edition.

- 2 types:
  1. Compact (dense)
  2. Trabecular (spongy; light)

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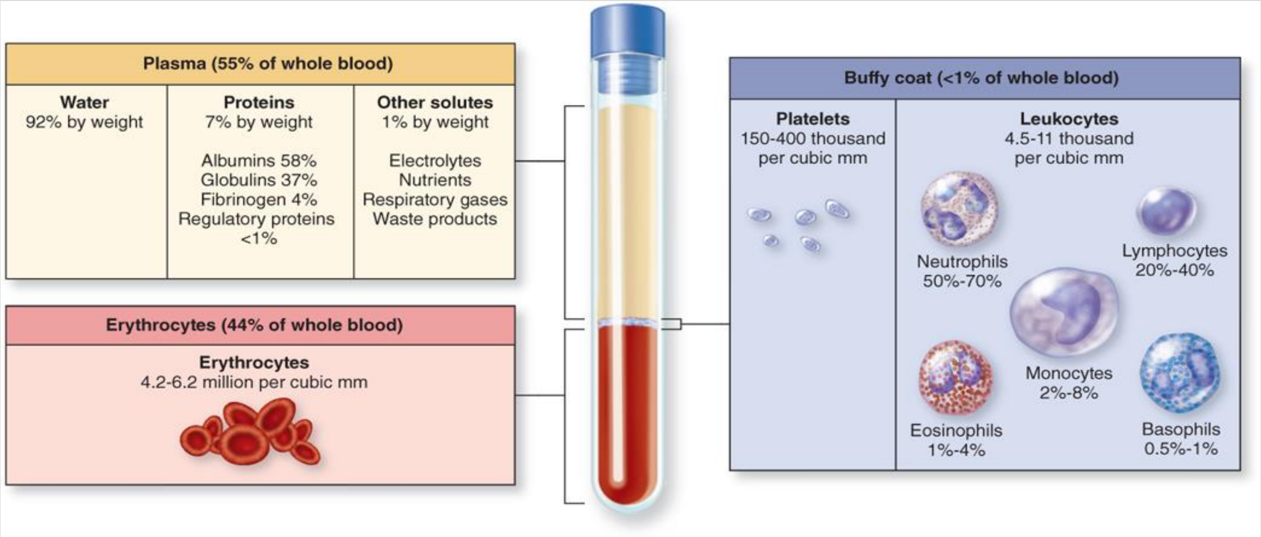
- Transport: O_2, CO_2, metabolites, hormones
- Heat distribution
- Regulation of body temperature
- Maintenance of acid-base & osmotic balance

Blood



Source: Anthony L. Mescher: Junqueira's Basic Histology: Text and Atlas, 15th Edition.

Muscle Tissue

- Function: Movement & heat production
- Types
  1. Skeletal
  2. Cardiac
  3. Smooth

Skeletal Muscle

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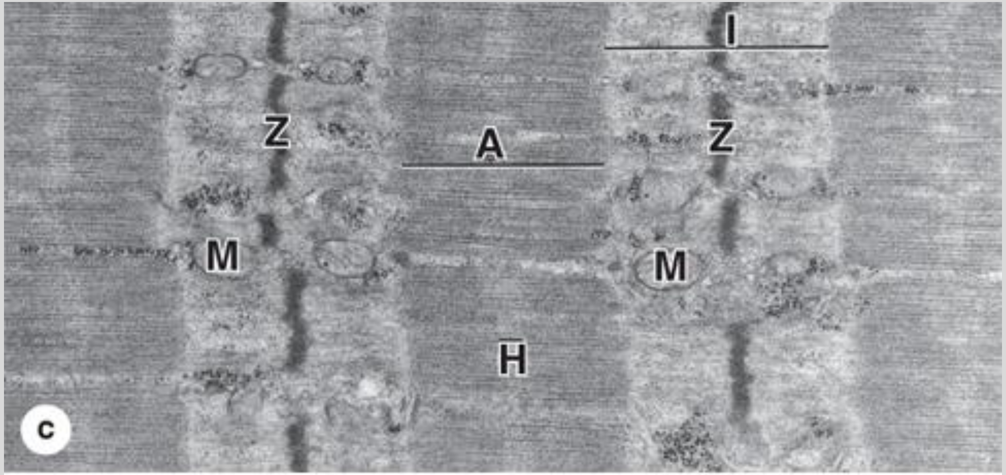
- Attached to bones
- Long, multinucleated cylindrical cells
- Mostly voluntary
- Striated
- Sarcomere



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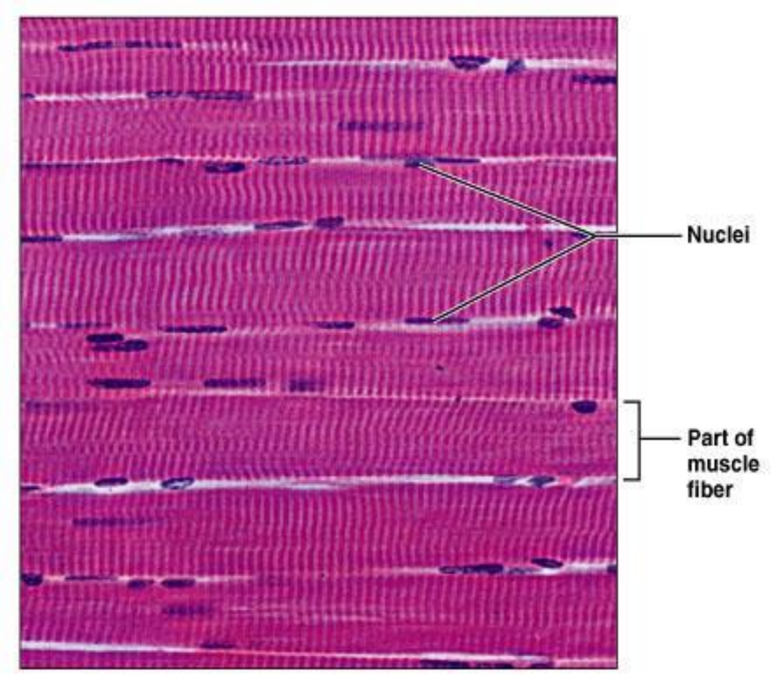
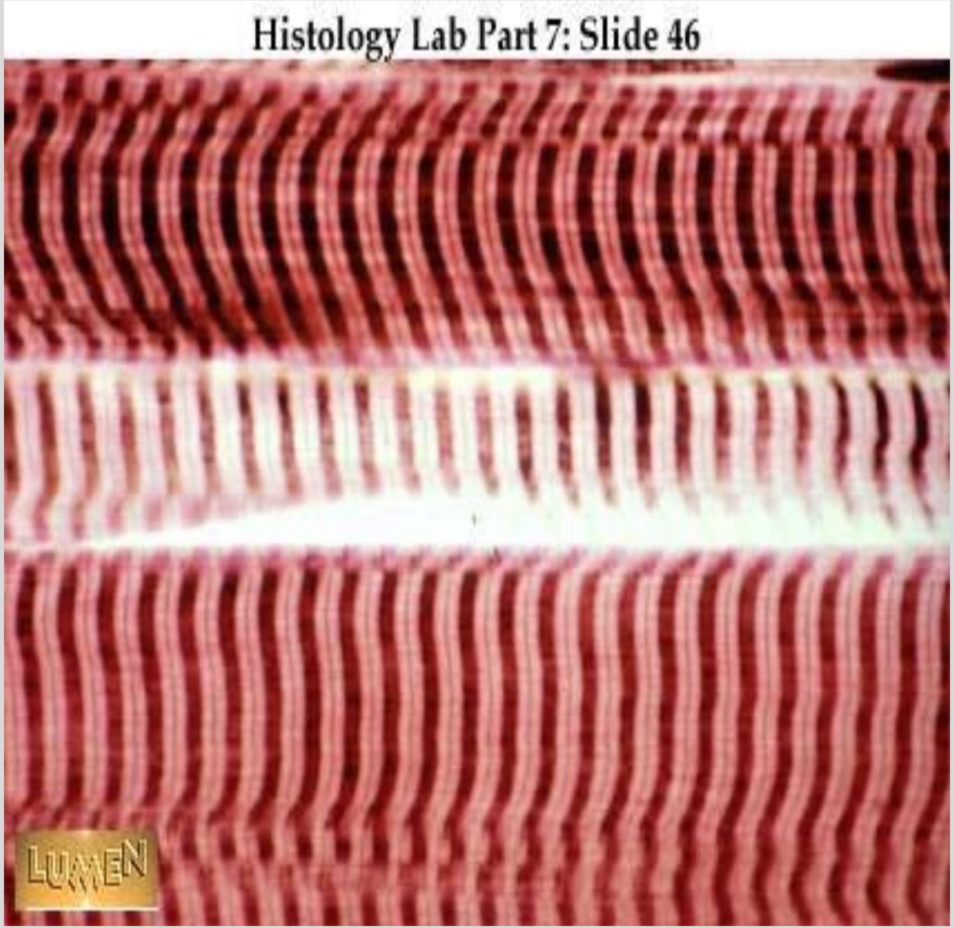


Skeletal muscle



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Cardiac Muscle

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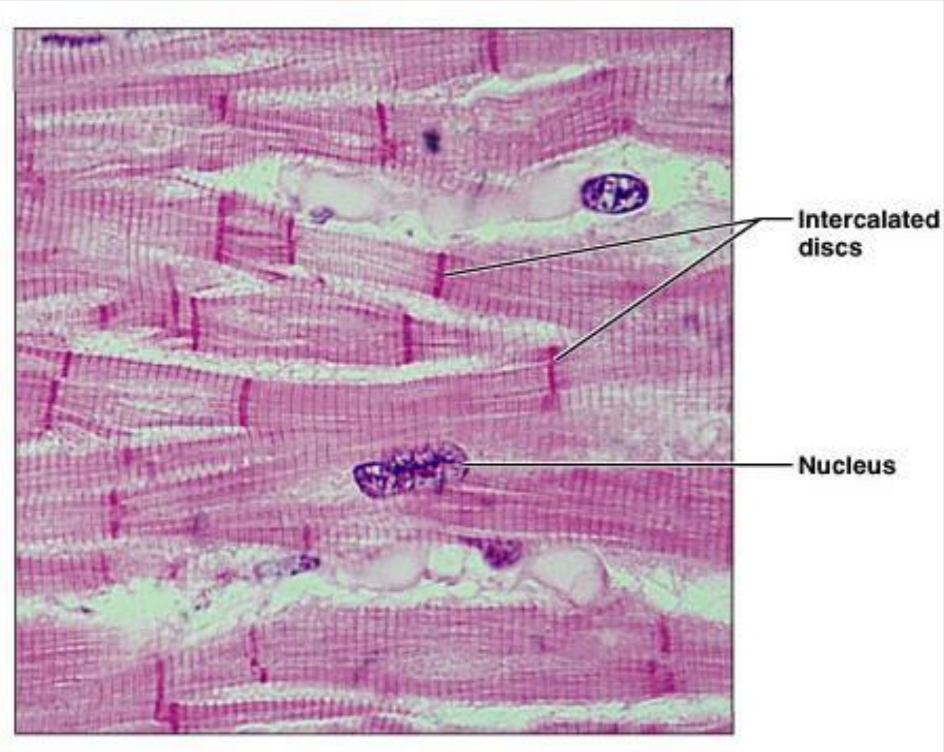
- Uni or bi-nucleated, branching cells (myocardium)
- Striated
- Intercalated discs
- Involuntary
- Autonomous



Smooth Muscle

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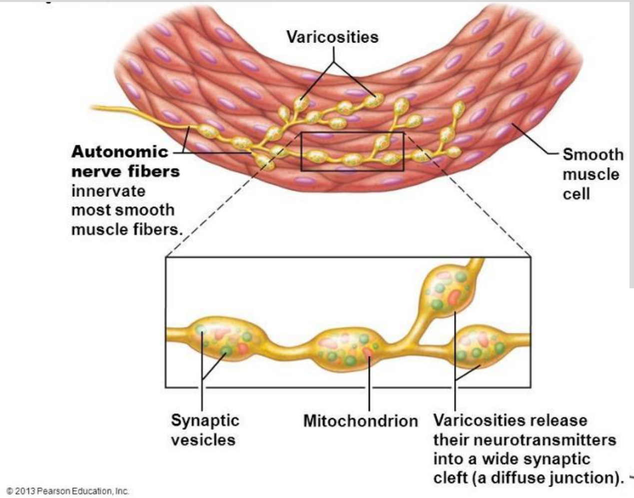
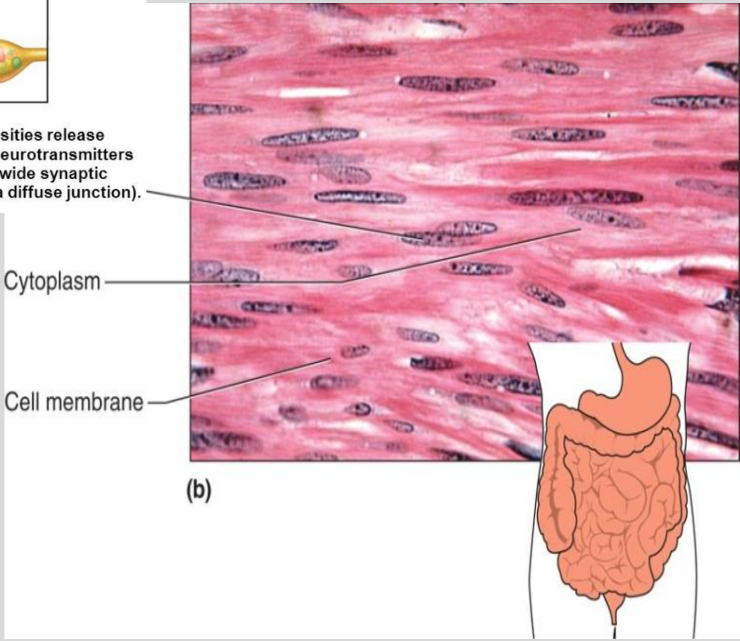
- Spindle-shaped cells with central nuclei
- No striations (no sarcomeres)
- Involuntary
- Autonomous
- Walls of hollow organs & blood vessels
- Neuromuscular junction is distinctive (varicosities)





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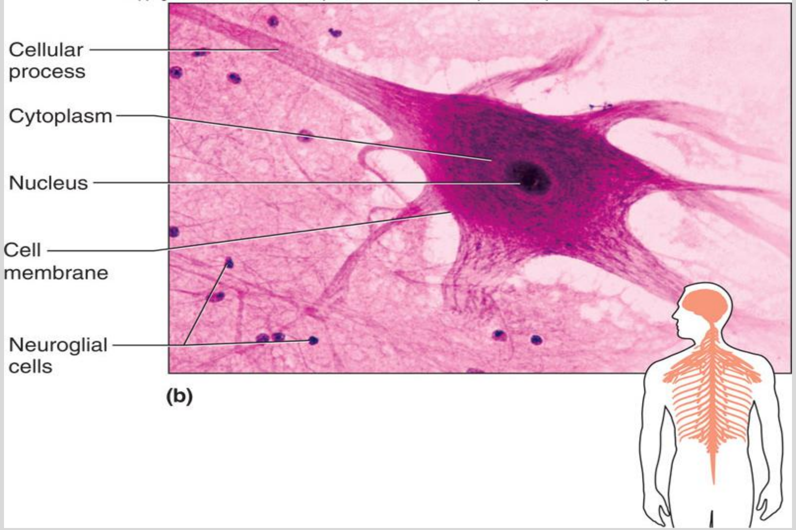

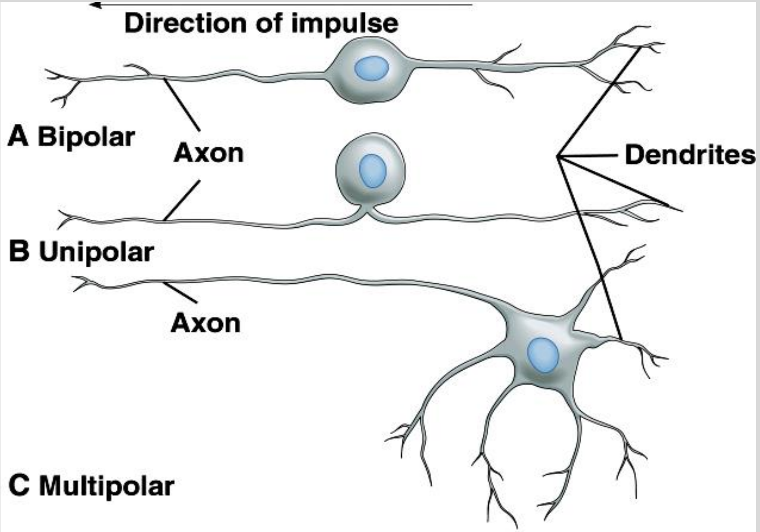

Nervous Tissue

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- Neurons & neuroglial cells
  a) Neurons:
     i. Processes: axons & dendrites
     ii. Uni, bi & multipolar
     iii. Sensory, motor & interneurons
  b) Glial cells:
     i. CNS: Astrocytes, oligodendrocytes, microglia & ependymal cells
     ii. PNS: Schwann & satellite cells
- CNS: brain & spinal cord
- PNS: cranial & spinal nerves

Tissues throughout life

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- Early on – Gastrulation
  a) Most important time in life!!
     a) When tissues differentiate – problems here affect development
- At end of second month of development
  a) Primary tissue types have appeared
  b) Major organs are in place
- Adulthood
  a) Only a few tissues regenerate
  b) Many tissues still retain populations of stem cells
- With increasing age:
  a) Epithelia thin
  b) Collagen decreases
  c) Bones, muscles, & nervous tissue begin to atrophy
  d) Poor nutrition & poor circulation – poor health of tissues
  e) Increased chance of developing cancer

Histogenesis

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- Trilaminar embryo – ectoderm, mesoderm, & endoderm
- Ectoderm – outermost layer
  1) Surface ectoderm: epidermis & derivatives, cornea, lens epithelia, enamel, internal ear, adenohypophysis, mucosa of oral & anal cavities
  2) Neuroectoderm: CNS & PNS
- Mesoderm – middle layer
  1) All connective tissue
  2) All muscle tissue
  3) Heart, blood vessels, & lymphatics
  4) Spleen
  5) Kidneys
  6) Gonads
  7) Mesothelium
  8) Adrenal cortex

Histogenesis

- Endoderm – innermost layer
  1) Alimentary canal epithelium
  2) Liver, pancreas, & gallbladder
  3) Lining of Urinary bladder/most of urethra
  4) Respiratory system epithelium
  5) Thyroid, parathyroid, & thymus
  6) Tonsils
  7) Epithelium of tympanic cavity & auditory tubes

References

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- Core References:

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  - Ch. 4 & 5 in Mescher AL. eds. 2018. **Junqueira's Basic Histology: Text & Atlas**, 15th ed. New York, NY: McGraw-Hill publication.
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